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22ESC143/243

VTU.REG.NO

B.E. First/Second Semester Examination – January 2024

(Summer Term)

Introduction to Electronics Engineering

(Common to CSE,CSD, ISE and AIML)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each Module.

Module I

1. a) With neat block diagram summarize the working of DC power supply and also mention the principle components used in each block. **(06 Marks)**
   b) With relevant figure and waveform, explain working of full wave bridge rectifier . Mention its equations. **(10 Marks)**
   c) With neat diagram , explain the function of reservoir and smoothing circuits. **(04 Marks)**

2. a) With neat diagram explain the working of single stage CE amplifier with negative feedback.. **(08 Marks)**
   b) An AC supply of 230 V is applied to FWR through a transformer of turns ratio 10:1. Find i) DC output power ii) AC input power ii) Efficiency. Assume ideal diodes **(05 Marks)**
   c) What are multi-stage amplifiers? What are the different methods used for interstage coupling? **(07 Marks)**

Module II

3. a) With neat circuit diagram, derivation, waveform, explain operation of op-amp as i) Inverting summer ii) Subtractor. **(08 Marks)**
   b) Compare characteristic of an ideal and practical op-amp. **(07 Marks)**
   c) Describe the working of a single stage astable multivibrator using an op-amp. **(05 Marks)**

4. a) Mention the condition of sustained oscillations. Determine the frequency of oscillations of a three stage ladder network in which C = 10nF and R = 10kΩ **(05 Marks)**
   b) With a neat circuit diagram, explain Wien bridge oscillator. **(10 Marks)**
   c) Explain the concept of virtual ground in op-amp. **(05 Marks)**

Module III

5. a) Convert :
   i) $(14.625)_{10} = (?)_2 = (?)_{16}$ ii) $(6327.45)_8 = (?)_{10} = (?)_{16}$
   iii) $(101111.101)_2 = (?)_8 = (?)_{10}$ iv) $(AF.2F)_{16} = (?)_{10} = (?)_8$. **(08 Marks)**
   b) Implement half adder using NAND and NOR gates. **(04 Marks)**
   c) Subtract suing 2's complement method: i) -56 and -65 ii) 110110101 from 111001010. **(08 Marks)**

6. a) Simplify the following functions::
   i) $f_1 = (\overline{A+B})(\overline{A}+\overline{B})(\overline{B}+\overline{C})$ and ii) $f_2 = AB + ABC + \overline{A}B + A\overline{B}C$. **(06 Marks)**
   b) Implement 2-input Ex-NOR gate using i) 2-input NAND gate ii) 2-input NOR gate. **(06 Marks)**
   c) With the help of truth table, explain the operation of Full adder with its circuit diagram and reduce the expression for Sum and carry & implement using basic gates. **(08 Marks)**

Module IV

7. a) Compare Embedded systems and General computing systems also provide the applications of Embedded systems. **(08 Marks)**
   b) Write a note on classification of Embedded systems. **(07 Marks)**
   c) Explain the operation of an LED with neat diagram . Mention its applications. **(05 Marks)**

8. a) What is an embedded system and brief about the different elements of an embedded system. (05 Marks)
   b) Explain the different configurations of 7 – segment LED display. (05 Marks)
   c) Differentiate i) RISC vs CISC processors ii) Microprocessor vs Microcontroller. (10 Marks)

Module V

9. a) With a neat block diagram , explain the basic communication system. (10 Marks)
   b) Explain digital modulation techniques. (10 Marks)

10. a) With neat diagram explain amplitude modulation, frequency modulation and phase modulation. (10 Marks)
    b) Explain different types of radio wave propagation with a neat diagram. (05 Marks)
    c) Define and explain SNR, noise figure and channel types. (05 Marks)